

Exercise Sheet 1

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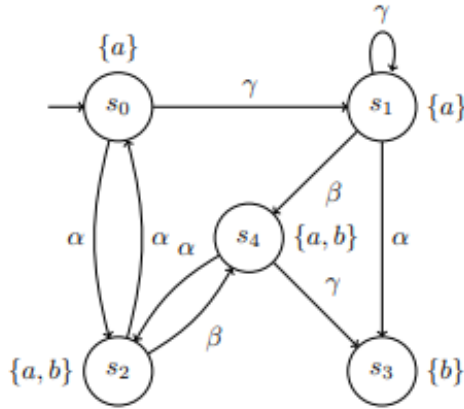
1 Exercise 1

We call a transition system $TS = (S, Act, \rightarrow, S_0, AP, L)$

- action-deterministic if $|S_0| \leq 1$ and $|Post(s, \alpha)| \leq 1$ for all $s \in S$ and $\alpha \in Act$, and
- AP-deterministic if $|S_0| \leq 1$ and $|Post(s) \cap \{s' \in S \mid L(s') = A\}| \leq 1$ for all $s \in S$ and $A \in 2^{AP}$,

where $Post(s, \alpha) = \{s' \in S \mid \exists (s, \alpha, s') \in \rightarrow\}$ and $Post(s) = \cup_{\alpha \in Act} Post(s, \alpha)$

Let the transition system TS_1 be as follows.



- Give the formal definition of TS_1
- Specify a finite and an infinite execution of TS_1
- Decide whether TS_1 is (i) AP-deterministic, and/or (ii) action-deterministic. Justify your answer.

Solution

(a) The formal definition of TS_1 :

- State Space: $S = \{s_0, s_1, s_2, s_3, s_4\}$
- Action: $Act = \{\alpha, \beta, \gamma\}$
- Transition relation:

$$\rightarrow = \{(s_0, \gamma, s_1), (s_0, \alpha, s_2), (s_1, \gamma, s_1), (s_1, \beta, s_4), (s_1, \alpha, s_3), (s_2, \alpha, s_0), \\ (s_2, \beta, s_4), (s_4, \alpha, s_2), (s_4, \gamma, s_3)\}$$

- Initial State: $S_0 = s_0$
- Atomic Propositions: $AP = \{a, b\}$
- Labeling Function:

$$L(s) = \begin{cases} \{a\} & \text{if } s \in \{s_0, s_1\} \\ \{a, b\} & \text{if } s \in \{s_2, s_4\} \\ \{b\} & \text{if } s = s_3 \end{cases}$$

(b) A finite execution of TS_1 :

$$s_0 \xrightarrow{\alpha} s_2 \xrightarrow{\beta} s_4 \xrightarrow{\gamma} s_3$$

An infinite execution of TS_1 :

$$(s_0 \xrightarrow{\alpha} s_2 \xrightarrow{\alpha} s_0)^\omega$$

(c) Based on what has been given, we have:

- $\forall s \in S, \forall \alpha \in Act, \text{if } (|S_0| \leq 1) \wedge (|Post(s, \alpha)| \leq 1) \implies \text{Action-deterministic (1)}$
- $\forall s \in S, A \in 2^{AP}, \text{if } (|S_0| \leq 1) \wedge (|Post(s) \cap \{s' \in S \mid L(s') = A\}| \leq 1) \implies \text{AP-deterministic (2)}$

Because $|S_0| = 1 \implies |S_0| \leq 1 = \text{True (3)}$

- Check TS_1 is AP-deterministic

2^{AP} is the set of all subsets of AP

$$2^{AP} = \{\phi, \{a\}, \{b\}, \{a, b\}\}$$

From (2), (3) $\implies$

For TS_1 is AP-deterministic, $|Post(s) \cap \{s' \in S \mid L(s') = A\}| \leq 1$ must be True

- With $s = s_0$:

$$Post(s_0) = \{s_1, s_2\}$$

- With $A = \phi$

$$\{s' \in S \mid L(s') = A\} = \{\}$$

$$\implies |Post(s) \cap \{s' \in S \mid L(s') = A\}| = |\{\}| = 0 \leq 1 \text{ (satisfied)}$$

- With $A = \{a\}$

$$\{s' \in S \mid L(s') = A\} = \{s_0, s_1\}$$

$$\implies |Post(s) \cap \{s' \in S \mid L(s') = A\}| = |\{s_1\}| = 1 \leq 1 \text{ (satisfied)}$$

- With $A = \{b\}$

$$\{s' \in S \mid L(s') = A\} = \{s_3\}$$

$$\implies |Post(s) \cap \{s' \in S \mid L(s') = A\}| = |\{\}| = 0 \leq 1 \text{ (satisfied)}$$

- With $A = \{a, b\}$

$$\{s' \in S \mid L(s') = A\} = \{s_2, s_4\}$$

$$\implies |Post(s) \cap \{s' \in S \mid L(s') = A\}| = |\{s_2\}| = 1 \leq 1 \text{ (satisfied)}$$

- With $s = s_1$

$$Post(s_1) = \{s_3, s_4\}$$

- With $A = \phi$

$$\{s' \in S \mid L(s') = A\} = \{\}$$

$$\implies |Post(s) \cap \{s' \in S \mid L(s') = A\}| = |\{\}| = 0 \leq 1 \text{ (satisfied)}$$

- With $A = \{a\}$

$$\{s' \in S \mid L(s') = A\} = \{s_0, s_1\}$$

$$\implies |Post(s) \cap \{s' \in S \mid L(s') = A\}| = |\{\}| = 0 \leq 1 \text{ (satisfied)}$$

- With $A = \{b\}$

$$\{s' \in S \mid L(s') = A\} = \{s_3\}$$

$$\implies |Post(s) \cap \{s' \in S \mid L(s') = A\}| = |\{s_3\}| = 1 \leq 1 \text{ (satisfied)}$$
- With $A = \{a, b\}$

$$\{s' \in S \mid L(s') = A\} = \{s_2, s_4\}$$

$$\implies |Post(s) \cap \{s' \in S \mid L(s') = A\}| = |\{s_4\}| = 1 \leq 1 \text{ (satisfied)}$$
- With $s = s_2$

$$Post(s_2) = \{s_0, s_4\}$$
 - With $A = \phi$

$$\{s' \in S \mid L(s') = A\} = \{\}$$

$$\implies |Post(s) \cap \{s' \in S \mid L(s') = A\}| = |\{\}| = 0 \leq 1 \text{ (satisfied)}$$
 - With $A = \{a\}$

$$\{s' \in S \mid L(s') = A\} = \{s_0, s_1\}$$

$$\implies |Post(s) \cap \{s' \in S \mid L(s') = A\}| = |\{s_0\}| = 1 \leq 1 \text{ (satisfied)}$$
 - With $A = \{b\}$

$$\{s' \in S \mid L(s') = A\} = \{s_3\}$$

$$\implies |Post(s) \cap \{s' \in S \mid L(s') = A\}| = |\{\}| = 0 \leq 1 \text{ (satisfied)}$$
 - With $A = \{a, b\}$

$$\{s' \in S \mid L(s') = A\} = \{s_2, s_4\}$$

$$\implies |Post(s) \cap \{s' \in S \mid L(s') = A\}| = |\{s_4\}| = 1 \leq 1 \text{ (satisfied)}$$
- With $s = s_3$

$$Post(s_3) = \{\}$$
 - With $A = \phi$

$$\{s' \in S \mid L(s') = A\} = \{\}$$

$$\implies |Post(s) \cap \{s' \in S \mid L(s') = A\}| = |\{\}| = 0 \leq 1 \text{ (satisfied)}$$
 - With $A = \{a\}$

$$\{s' \in S \mid L(s') = A\} = \{s_0, s_1\}$$

$$\implies |Post(s) \cap \{s' \in S \mid L(s') = A\}| = |\{\}| = 0 \leq 1 \text{ (satisfied)}$$
 - With $A = \{b\}$

$$\{s' \in S \mid L(s') = A\} = \{s_3\}$$

- $\implies |Post(s) \cap \{s' \in S \mid L(s') = A\}| = |\{\}| = 0 \leq 1$ (satisfied)
- With $A = \{a, b\}$
 - $\{s' \in S \mid L(s') = A\} = \{s_2, s_4\}$
 - $\implies |Post(s) \cap \{s' \in S \mid L(s') = A\}| = |\{\}| = 0 \leq 1$ (satisfied)
- With $s = s_4$
 - $Post(s_4) = \{s_2, s_3\}$
 - With $A = \phi$
 - $\{s' \in S \mid L(s') = A\} = \{\}$
 - $\implies |Post(s) \cap \{s' \in S \mid L(s') = A\}| = |\{\}| = 0 \leq 1$ (satisfied)
 - With $A = \{a\}$
 - $\{s' \in S \mid L(s') = A\} = \{s_0, s_1\}$
 - $\implies |Post(s) \cap \{s' \in S \mid L(s') = A\}| = |\{\}| = 0 \leq 1$ (satisfied)
 - With $A = \{b\}$
 - $\{s' \in S \mid L(s') = A\} = \{s_3\}$
 - $\implies |Post(s) \cap \{s' \in S \mid L(s') = A\}| = |\{s_3\}| = 1 \leq 1$ (satisfied)
 - With $A = \{a, b\}$
 - $\{s' \in S \mid L(s') = A\} = \{s_2, s_4\}$
 - $\implies |Post(s) \cap \{s' \in S \mid L(s') = A\}| = |\{s_2\}| = 1 \leq 1$ (satisfied)

$$\implies |Post(s) \cap \{s' \in S \mid L(s') = A\}| \leq 1 = \text{True} \quad (4)$$

From (3), (4) $\implies TS_1$ is AP-deterministic

- Check TS_1 is Action-deterministic

From (1) and (3) $\implies$ For TS_1 is action-deterministic, $|Post(s, \alpha)| \leq 1$ must be True

- Case $s = s_0$:
 - $Post(s_0, \gamma) = \{s_1\}, Post(s_0, \alpha) = \{s_2\}$ (satisfied)
- Case $s = s_1$:
 - $Post(s_1, \beta) = \{s_4\}, Post(s_1, \alpha) = \{s_3\}$ (satisfied)
- Case $s = s_2$:
 - $Post(s_2, \beta) = \{s_4\}, Post(s_2, \alpha) = \{s_0\}$ (satisfied)

- Case $s = s_3$: (satisfied because there are no way to reach to any state)
- Case $s = s_4$:
 $Post(s_4, \alpha) = \{s_2\}, Post(s_4, \gamma) = \{s_3\}$ (satisfied)

$$\implies |Post(s, \alpha)| \leq 1 = \text{True} \quad (5)$$

From (3), (5) $\implies TS_1$ is action-deterministic

2 Exercise 2

Consider the following sequential hardware circuit in C